Arizona Solar-Powered EV Fast Charging

AZ Solar EV DCFC - Hybrid ML + Optimization
IISE Hackathon 2025

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Problem & Objectives

Place a finite number of solar-augmented DC fast charging (DCFC) sites across Arizona to:

1. Maximize access and equity for current EV drivers
2. Minimize lifecycle cost (including grid energy when solar is insufficient)
3. Ensure reliability under Arizona's extreme heat
4. Comply with corridor coverage expectations while avoiding congestion/safety pitfalls

Why This Matters (Challenge & Impact)

Range Anxiety: Only 45% of rural U.S. counties have a fast charger

Extreme Heat: 110°F+ summers degrade equipment & charging efficiency

Grid Constraints: Single 150 kW charger = entire neighborhood's load

Equity Gaps: Underserved communities lack charging access

Key Decisions to Optimize

- Where to place stations (locations)
- How many ports per station (capacity)
- How much solar to install (PV sizing)
- Whether to include battery storage (resilience)

Our Solution - Hybrid ML + Optimization

ML Module
Predict Demand → MILP Module
Optimize Sites

Stage 1: Machine Learning Demand Prediction

- Ensemble of 3 Models: XGBoost + LightGBM + CatBoost
- Predicts: Expected daily energy demand (kWh) at each candidate site
- Explainable: SHAP values show which factors drive predictions

Stage 2: Mixed-Integer Linear Programming (MILP)

- Framework: Pyomo with HiGHS solver
- Multi-objective: Maximize utility while minimizing cost and risk
- Constraints: Budget, spacing, coverage requirements

Alternative	Why not chosen
Manual Scoring	Subjective weights, misses nonlinear relationships
Pure Heuristics	No optimality guarantee, hard to tune
Full Simulation/RL	Too complex for 24-hour hackathon

Factors to consider

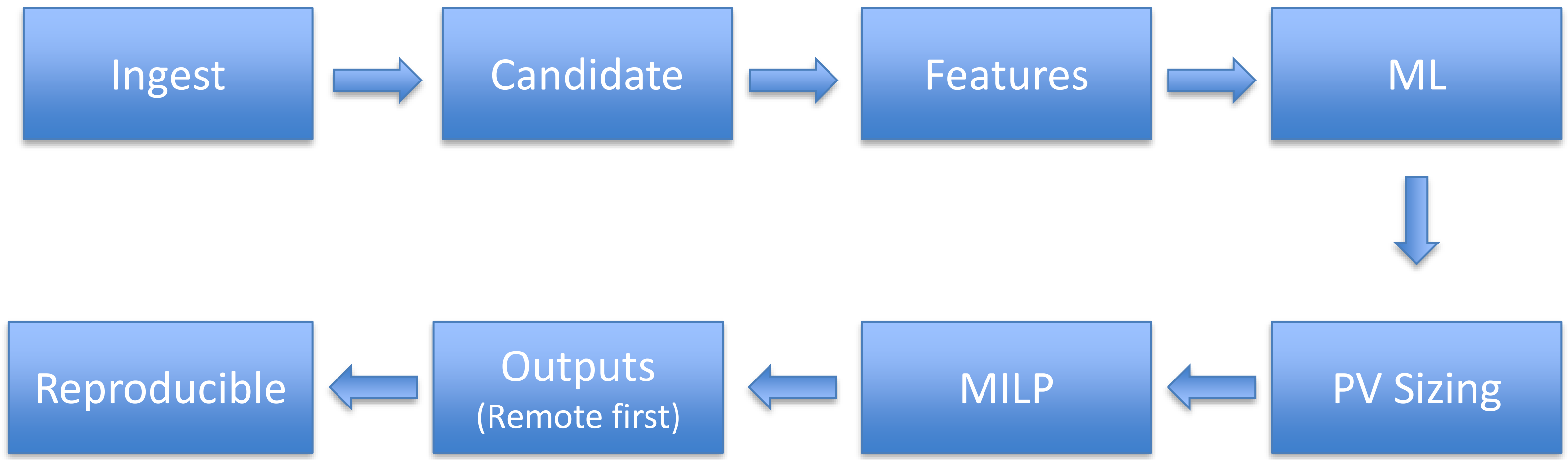
Category	Factors
Demand & Accessibility	Traffic volume (AADT), Population density, Trip attractors, Equity need
Spacing & Corridors	Max distance between chargers, Detour from highway, EV range feasibility
Site Feasibility	Parcel availability, Zoning compatibility, Permitting complexity
Safety & Security	Crime risk, Lighting/surveillance, Flood/fire hazards
Grid & Interconnection	Utility territory (APS/SRP), Hosting capacity, Tariff structure
Solar Resource	Annual irradiance, Ambient temperature, Thermal derating
Energy Balance	PV sizing vs load, Battery for peak shaving, Grid draw profile
Costs & Economics	CapEx components, OpEx rates, LCOC, Budget constraints
Operations & Maintenance	Heat/dust maintenance, Parts replacement, Uptime SLAs
Traffic & UX	Queueing risk, Parking layout, Driver amenities
Policy Compliance	NEVI standards, Electrical codes, Accessibility
Environmental Impact	Noise/visual, Heat-island mitigation, Stormwater
Resilience	Heat design margins, Backup power, Site redundancy

Public Datasets

Dataset	What we use it for	Source/Access
ADOT AADT	Traffic volumes; candidate seeds; demand features	ArcGIS FeatureServer (public)
NREL AFDC	Existing EV stations; nearest-DCFC competition	NREL API (key)
FEMA NFHL	Flood-risk screening; safety penalties	MapServer (public)
Census ACS	Demographics (ZCTA): renters, income, vehicles/HH	Census API (optional key)
EPA EJSCREEN*	Equity indicators for fairness objectives	CSV download (public)
Park & Ride	Public hosts with safe access	ArcGIS FeatureServer (public)

City Block is too noisy for statewide datasets while County is too coarse for siting decisions. So, ZCTA (ZIP) is a sweet spot which is linkable to ACS, compatible with corridor

Pipeline Overview



Optimization Model & Difficulty

MILP Problem Formulation

Capacitated Facility Location / P-median hybrid Solver: HiGHS (open-source, branch-and-cut) Time Limit: 10 minutes with 1% optimality gap

For 500 candidates: 2,500+ decision variables!

Multi-Objective Function To Maximize

w_util × Σ(open[i] × pred_daily_kwh[i]) # Utilization

+ w_equity × Σ(open[i] × equity_score[i]) # Equity

- w_safety × Σ(open[i] × safety_penalty[i]) # Safety risk

- w_grid × Σ(open[i] × grid_conflict[i]) # Grid conflict

- w_cost × (Total Cost / 1,000,000) # Normalized cost

Continued...

Constraint	Formula	Default
Budget	Σ CapEx $\le$ B	$15,000,000
Site Count	Σ open[i] $\le$ N	40 sites
Min Spacing	dist(i,j) $\ge$ D if both open	50 km
Min Ports	ports[i] $\ge 4 \times$ open[i]	4 ports
Max Ports	ports[i] $\le 8 \times$ open[i]	8 ports
Min PV	pv_kw[i] $\ge 50 \times$ open[i]	50 kW
Max PV	pv_kw[i] $\le 300 \times$ open[i]	300 kW
Max Storage	storage_kwh[i] $\le 500 \times$ open[i]	500 kWh

What Makes This Difficult

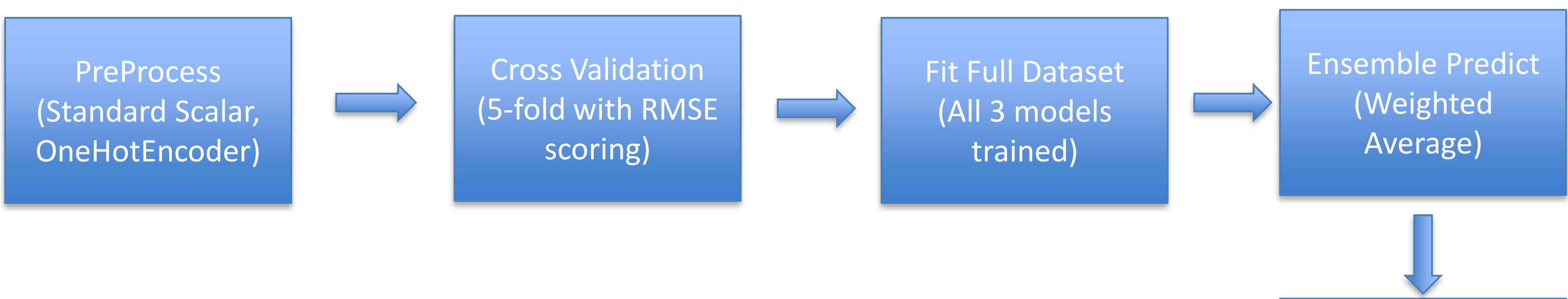
- Mixed-Integer: Binary + Integer + Continuous variables
- Spatial Constraints: Pairwise spacing creates $O(n^2)$ constraints
- Multiple Objectives: Balancing competing goals
- Nonlinear Relationships: Linearized for MILP feasibility
- Large Scale: 500+ candidates × 5 variables = 2500+ decision vars

Machine Learning Models

Ensemble Architecture
Three Gradient Boosting Models with equal-weighted blending:

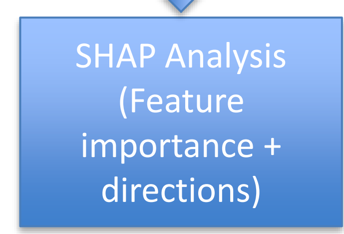
Model: Strength
XGBoost: Fast, robust to outliers
LightGBM: Handles large datasets
CatBoost: Native category handling

Training Pipeline



SHAP (SHapley Additive exPlanations) provides:

- **Feature Importance**: Which inputs matter most
- **Direction**: Positive or negative contribution
- **Magnitude**: Strength of effect



Results & Economic Analysis

Site Selection:

- 15-30 optimally placed stations
- Geographic coverage across Arizona's major corridors
- Mix of high-demand and underserved equity locations

Portfolio Economic Summary

Metric	Expected Range
Total CapEx	$12-15M
Net CapEx (after incentives)	$3-5M
Annual OpEx	$1.5-2M
Annual Revenue	$2-3M
Portfolio NPV	$5-10M

Per-Site Breakdown

Component	Typical Value
Site Prep + Civil	$125,000
Interconnection	$150,000
4-8 DC Chargers	$260,000-520,000
PV System (100 kW)	$160,000
Battery (200 kWh)	$120,000
Total CapEx	$815,000-1,075,000

Incentives applied: Federal ITC (30% of PV) and NEVI Grant (80% of CapEx) resulting in 80% reduction

Summary

- Hybrid ML + Optimization: Data-driven predictions + mathematically optimal selection
- 13 Factor Categories: Comprehensive consideration of all siting factors
- 6+ Public Data Sources: Grounded in real, accessible data
- Multi-Objective MILP: Balances utilization, equity, safety, and cost
- Explainable AI: SHAP values justify every decision
- Arizona-Specific: Heat derating, solar optimization, APS/SRP territories
- End-to-End Solution: From raw data to interactive dashboard

Resources:

https://elektraz-5wkjmdcnzbn9c7t7cuyjza.streamlit.app/

https://github.com/ak-asu/elektraz